Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/27/2013
PI Name:	Cannell, David S Ph.D.		
Project Title:	Gradient Driven Fluctuations-Phase 2		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106-9530	**Congressional District:**	23
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/01/2008	**End Date:**	08/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	Foton		
Flight Assignment:	Foton M3 NOTE: New end date is 8/31/2012 per NSSC information (Ed., 6/20/2012) NOTE: New end date is 2/28/2012 per NSSC information (Ed., 5/31/2011)		
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NNX08AE53G		
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Performance Goal Text:			

Task Description:	GRADFLEX, a co-operative effort between NASA, ESA, the University of Milan and the University of California at Santa Barbara (UCSB) was flown aboard the Foton-M3 satellite mission in September 2007. The apparatus applied various thermal gradients to both a sample of a single-component fluid (carbon disulphide) (SINGLE) and a 2-component mixture (polystyrene in toluene) (MIXTURE). Thermal fluctuations under these conditions are greatly enhanced both in amplitude and spatial extent. This effect occurs on Earth, but is limited by gravity, while in micro-gravity, the range of the fluctuations is predicted to be limited only by the sample dimension. A sensitive shadowgraph method capable of carrying out both static and dynamic light scattering at scattering angles as small as one thousandth of a degree was developed and used for this experiment. This method is now beginning to be applied in other areas, such as colloidal physics. The flight returned approximately 60 Gb of data images, while ground-based runs, completed using the same apparatus after flight, added another 400 Gb of data. This grant requests support for the sorting, decompression and analysis of this massive data set. Because the instrument behaved in an unexpected way when gradients were applied during flight (but not after it was safely back on Earth!), the question of how to analyze the flight data has no obvious a priori answer. Initial attempts have shown that unexplained spectral power is present for low wave vector, and this spectral power appears in the form of a narrow band, very low frequency, noise process. The relative contribution of this noise source varies with position in the cell, and from run to run. It is stronger with increasing gradient. The fact that it has a temporal power spectrum that differs markedly from that of the gradient-driven fluctuations offers the possibility that it can be separated via spectral analysis. Another complication that must be addressed is the role played by optical imperfections on the analysis of shadowgraph data. This unfortunate question has arisen because the sapphire window used in flight for SINGLE was not of the highest quality. It appears perfectly possible to expand the theory of the shadowgraph to include static imperfections, and this will be attempted before full data analysis is undertaken. There appear to be only relatively minor problems with both the static and dynamic data obtained on Earth after flight, and they should not be insurmountable. If the spectral method of analyzing the flight data is successful, the same method will be used for the ground-based data in order to be consistent. The questions to be answered from the analysis are: 1. Does the predicted spectral shape for micro-gravity describe the flight data well? 2. Is the mean-squared amplitude of the fluctuations S(q) on Earth, compatible with the predicted value, in absolute terms? If this question can be answered, even to within 10% accuracy, it will be an excellent test of theory. It would also represent an unprecedented technical achievement in the field of light scattering. 3. Does the removal of gravity, enhance the mean-squared amplitude of the fluctuations and alter the shape of S(q) to the extent and in the manner predicted? 4. Do the dynamics measured on Earth, show the same sort of minor differences from what is predicted as did the data obtained using the UCSB prototype instrument and which have already been published (Physical Review Letters, 100, 234502 (2008))?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Throughout the course of this work it has been necessary to constantly improve the experimental method of choice, namely the quantitative shadowgraph. This development has now reached the stage where it is clear that this method, if handled properly, represents a viable method of carrying out both dynamic and static light scattering at scattering angles so small as to defy the imagination of anyone who has worked in the field. Although it is likely to take some time for the knowledge we have gained to be disseminated, there is no reason this method cannot be added to the scattering arsenal. Furthermore, it can also be applied to X-rays, enabling one to extend the lower limit of X-ray scattering as well. The consequences of putting a new technique into the hands of researchers is difficult to asses in advance. Basically, however, this method will enable quantitative measurements of various systems having size ranges that are too large for existing techniques. Obvious examples are the crystallization process in colloidal systems, gelation processes, and aggregation and flocculation. Because scattering provides excellent statistical averages, the method should prove quite complementary to visualization techniques such as microscopy.
Task Progress:	Support from this grant enabled the analysis of all the flight and ground-based data to be carried out to a successful conclusion. Although it was never possible to identify the source of the very low frequency noise that visibly affected the flight images, it was possible to use spectral analysis in the time domain to separate the spectrum of the gradient-driven fluctuations from that of the noise, and by integrating the temporal spectrum of the fluctuations, obtain accurate results for the amplitude of the fluctuations as a function of their wavelength. It was also possible to extend the theory of the shadowgraph to include the effect of optical imperfections to lowest order, and this information was used in analyzing the data. Fortunately, the data analysis showed that even rather severe imperfections have relatively little effect on quantitative shadowgraph data, revealing another advantage of this new method. By replacing the sample by a calibration sample consisting of small (2 micron diameter) polystyrene latex spheres suspended in isopropyl alcohol (to protect the optical coatings in contact with the sample), after the flight, it was possible to obtain an absolute calibration of the entire system as a function of wave vector (inversely proportional to the wavelength). Because of this new calibration method, we were able to compare the experimental data with theory in absolute terms. By this is meant without adjusting any parameters, including the amplitude, something quite unusual in this area of research. The results of this flight experiment and the subsequent ground-based experiments were to show that current theoretical calculations for non-equilibrium fluctuations driven by an applied temperature gradient for a single-component fluid are accurate to within 15% or better. Given the assumptions made in the theory, not to mention the excess noise problem, this level of agreement is excellent. The one area in which is would appear that theory and experiment do not fully agree is the presence of some degree of excess spectral power at relatively low frequencies in ground-based experiments. No explanation exists for this relatively small effect.

	The results of the experiments carried out on the single-component sample were published in the prestigious journal Physical Review Letters (Physical Review Letters. 2011 Jun 17;106(24):244502). In addition to the work on the single-component fluid supported directly by this grant, the team in Milan completed an analysis of the flight data for a mixture of low molecular weight polystyrene dissolved in toluene. No attempt was made to provide an absolute comparison of the fluctuation amplitude with theory, but the dependence on wavelength was found to be excellent. This work was supported by the current grant to the extent that the PI was deeply involved in the collaboration and the paper writing. This work was published as a Nature Communication (Nature Communications., 2011;2:290). (Note publications reported in previous project's Bibliography.)
Bibliography Type:	Description: (Last Updated: 02/27/2013)